z-transform and ROC

Cheat sheet · Day 13 · Thu 5 Nov · pack D7

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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z-TRANSFORM (L07)   X(z) = sum x(k) z^-k ;  coeff of z^-n is x(n).  ALWAYS give ROC.
ROC: ring about 0, contains no poles, bounded by poles.
  finite: all z (drop 0 if n>0 samples, drop inf if n<0 samples)
  causal: |z| > outermost pole      anti-causal: |z| < innermost pole
  two-sided: ring between poles (empty -> no transform)
PAIRS: delta(n)->1 ; delta(n-d)->z^-d
  a^n u(n)       -> 1/(1-a z^-1), |z|>|a|
  -a^n u(-n-1)   -> 1/(1-a z^-1), |z|<|a|   (SAME X, different ROC; watch sign)
  n a^n u(n)     -> a z^-1/(1-a z^-1)^2, |z|>|a|
  a^n cos(w0 n)u(n) -> (1-a cos w0 z^-1)/(1-2a cos w0 z^-1+a^2 z^-2), |z|>|a|
PROPERTIES: x(n-d)->z^-d X (x(n+4)->z^4 X) ; z0^n x->X(z/z0), |z0|R
  e^{jw0 n}x -> X(e^{-jw0} z) ; x(-n)->X(1/z), radii inverted & swapped
  x_(k)(n)->X(z^k), R^(1/k) ; n x(n) -> -z dX/dz ; conv -> product, ROC >= RxnRh
  IVT (causal only): x(0) = lim_{z->inf} X(z)
POLES/ZEROS: write in +ve powers of z. z^(N-M) factor -> zeros (N>M) or poles (M>N) a
  X(inf)=0 -> zero at inf ; X(inf)=inf -> pole at inf. #poles = #zeros incl 0, inf.
BLOCKS: cascade H1H2 ; parallel H1±H2 ; feedback G/(1+GF) (minus at adder), G/(1-GF)
  Causal? expand final H(z): any z^{+k} term -> h(-k)≠0 -> NOT causal.
DTFT: X(w) = X(z) at z=e^{jw}, only if ROC contains |z|=1. X(0)=X(z=1), X(pi)=X(z=-1)
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